IOSP ATSCIENCE WORKSHOP · LEIDEN

# NOIZETOYZ

## Hardware Handout

Three reference sheets for building and debugging this station's boards — pulled from real firmware and real hardware sessions, not generic datasheet redraws. Print double-sided; each section starts on its own page.

1. Board & sensor pinouts — every board wired into the synth
2. DHT22 / DS18B20 sensor wiring — the shared D5 pull-up circuit
3. OLED status icons — what the synth board's display is telling you

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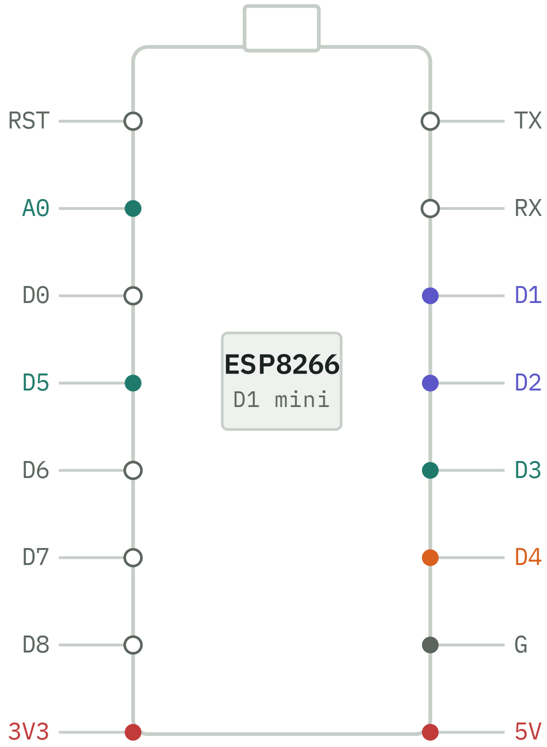
🎹🎶 NOIZETOYZ — HARDWARE REFERENCE

# Noizetoyz Pinouts

Real pin assignments for every board wired into the synth, pulled straight from this repo's firmware — not a generic datasheet redraw. **Copper** traces the actual audio signal across boards: GPIO out on the Wemos or UNO, in and back out the far side of the LM386.

## Wemos D1 mini

ESP8266 · esp_synth / bmp180 / note_player / dht22




D1/D2 pull double duty — I²C bus for the sensor and OLED sketches, plain buttons in the synth sketch. Same physical pins, different job per sketch.

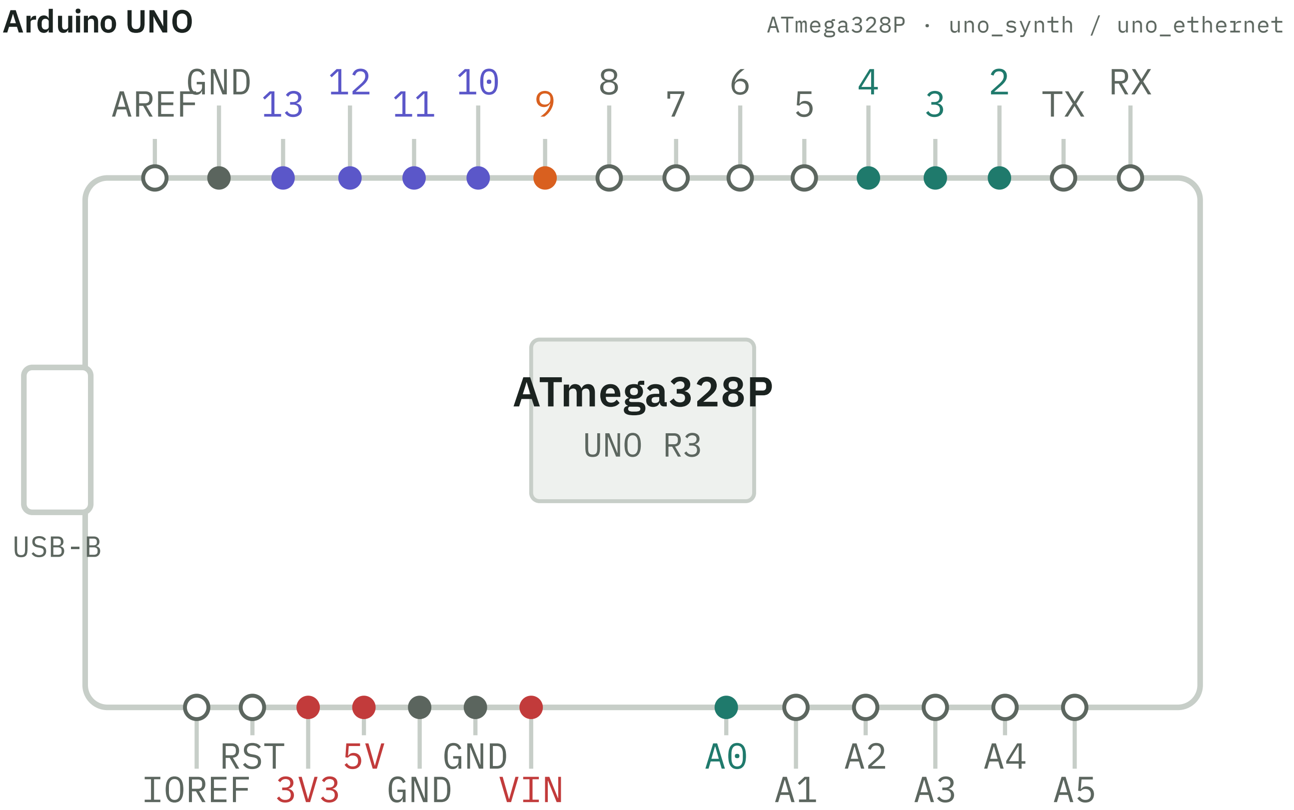
`esp_synth.ino` — D1/D2/D3 buttons, A0 pot (fx amount), D4 fixed Mozzi audio out.

`bmp180_wifi.ino` / `bmp180_smoke_test.ino` — D2 SDA, D1 SCL.

`esp_note_player.ino` — D4 audio out, D1/D2 I²C to the OLED shield.

`dht22_wifi.ino` / `ds18b20_test.ino` — D5 sensor data line (a separate board — see the wiring section).

## Arduino UNO




`uno_synth.ino` — pins 2/3/4 buttons, A0 pot, pin 9 Mozzi audio out (per Mozzi's own pin table).

`uno_ethernet_smoke_test.ino` — W5100-class shield on the standard SPI bus (11/12/13), CS on pin 10.

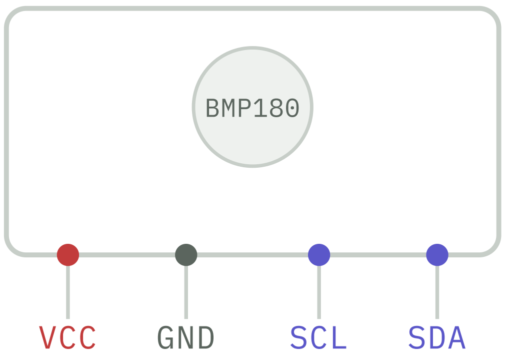
**LM386 amp module**

GS26347 · ×4 channels on one strip


`IN` takes the filtered PWM straight off the Wemos/UNO's audio pin; `OUT` drives a speaker. One strip carries four identical channels — enough for every hardware variant in the room at once.

## BMP180

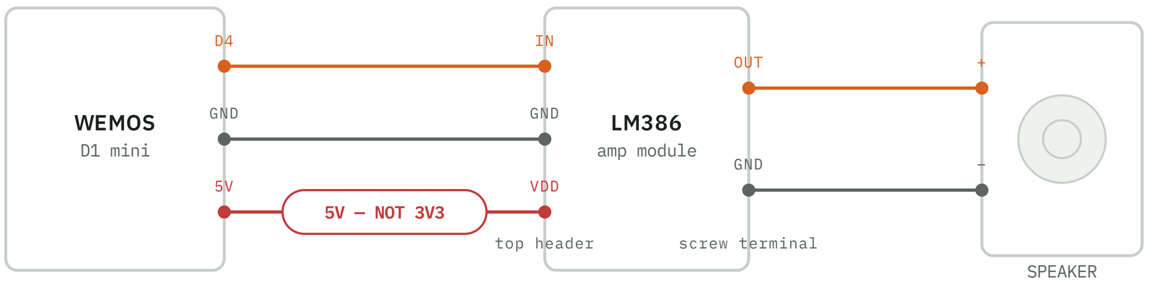
temp / pressure · I²C, addr 0x77




Same four pins regardless of which relay it feeds — live-data's `wifi_sensor_relay.py` or noizetoyz's `bmp180_wifi.ino`, both read it over I²C. Confirmed publishing live readings end-to-end 2026-09-04.

## Wiring: Wemos → amp → speaker

signal chain · 3 connections, 1 gotcha




Three wires, no more: `D4` + a `GND` off the Wemos into the amp module's top header `IN` + `GND`; the amp's bottom screw terminal `OUT` + `GND` straight to a speaker. The one gotcha is power — the amp's `VDD` comes off the Wemos's `5V` pin, **not** `3V3`. The regulated 3.3V rail only has ~200–300mA of headroom and is already feeding the ESP8266's own WiFi radio; the LM386 wants upward of 4V to behave, and sharing that rail risks brownouts or resets under load.

● power ● ground ● signal (button / pot / data) ● bus (I²C / SPI)

● audio path ○ unused here

Traced from this repo's firmware (`noizetoyz/firmware/`, `live-data/wifi_sensor_relay.py`) and real hardware photographed 2026-09-01 through 2026-09-04 — pin roles reflect what these sketches actually do.

SYNTH · LIVE-DATA CROSSOVER

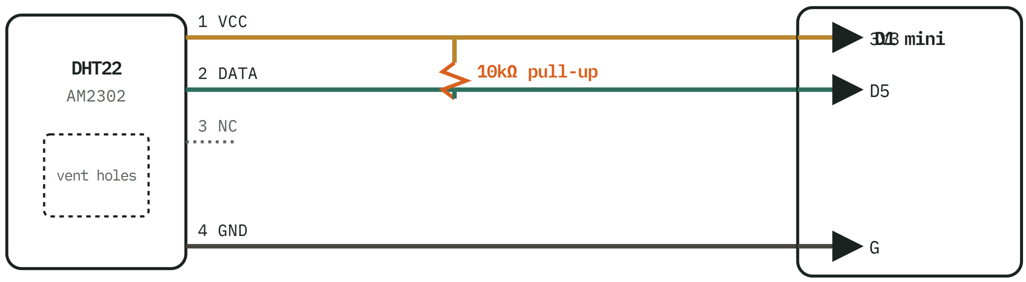
# DHT22 / DS18B20 sensor wiring

Both sensors share one circuit: D5 for data, 3V3 for power, a pull-up bridging the two — confirmed 2026-09-04 by swapping sensors on the exact same untouched wiring (this is exactly how a dead DHT22 unit was isolated from an otherwise healthy circuit). Only the physical pin order differs.

power data + pull-up ground

## DHT22 / AM2302

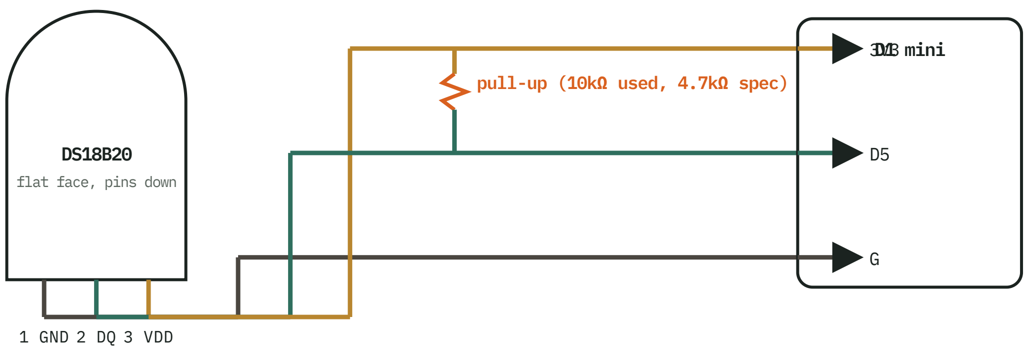
bare sensor on breadboard




VCC→3V3, GND→G, DATA→D5. Pin 3 (NC) is left unconnected. The 10kΩ pull-up bridges the VCC rail and the DATA line right at the sensor end — bare sensors like this one carry no onboard pull-up, unlike a breakout PCB.

## DS18B20

TO-92, flat face toward you, pins down




VDD→3V3, GND→G, DQ→D5 — same three D1 mini pins the DHT22 uses, same pull-up placement. Datasheet-spec pull-up is 4.7kΩ; the 10kΩ already on the breadboard from the DHT22 wiring works fine left in place.

#### WHY 3V3, NOT 5V

Both sensors are happy on 3.3–6V, but the ESP8266's GPIOs are not 5V-tolerant (~3.6V absolute max). 3V3 keeps DATA/DQ's HIGH level matched to the ESP8266's own logic level.

#### WHY D5 SPECIFICALLY

GPIO14 — one of the DHT library's own documented safe ESP8266 pins. D3/D4 (GPIO0/2) have boot-mode strapping behavior; D5 doesn't collide with I²C (D1/D2) or Mozzi's audio pin (D4).

#### WHY THE PULL-UP MATTERS

Both sensors use a single data line that must be pulled high when idle. No pull-up (or one bridging the wrong lines) reads as total silence — indistinguishable from a dead sensor without a cross-check.

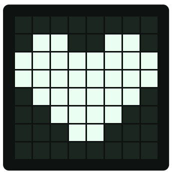
## What actually varies between them

Everything on the D1 mini side of both diagrams is identical — pins, pull-up placement, ground rail. All that changes is the sensor's own physical pin order. Swapping one sensor for the other is a drop-in test of the sensor alone — exactly how `firmware/esp-wifi/ds18b20_test/` isolated a dead DHT22 unit from an otherwise completely healthy circuit on 2026-09-04.

SYNTH · ESP_MULTI_SYNTH_OLED.INO

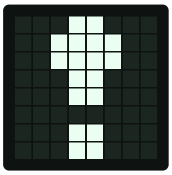
# Noizetoys status icons

8×8, one bit per pixel — the real constraint of the SSD1306 shield wired to every board. These four are flashed and confirmed on real hardware, right-aligned in the footer row of the synth board's display.



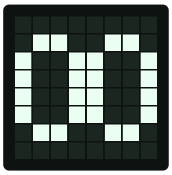
**Heart**

relay config fetched from real ATProto/HappyView



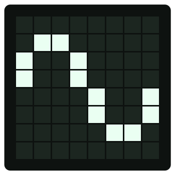
**Exclamation**

DEFAULT_RELAY_HOST fallback in use



**Link**

downlink connected



**Broken link**

downlink not connected

`Byte order matches PROGMEM arrays: one byte per row, MSB first, top to bottom. Right-aligned in the footer row, opaque-masked over the scrolling marquee text so it never shows through.`